PS NO. 52 | PROJECT REPORT

Digital Diary Notes Manager

A console-based diary program in Python that saves notes to a file

BACHELOR OF COMPUTER APPLICATIONS (BCA) - RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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PROBLEM STATEMENT
PS No. 52

PROGRAMME
BCA

1 Problem statement

For this project I had to make a diary program in **Python** that runs in the console. The user can add notes with a date, see all notes, search for a note using a keyword, and delete a note. All the notes are saved in a file called `diary.txt`, so they are still there after I close the program.

1. **Add a note** with a date
2. **View all** numbered notes
3. **Search** by keyword
4. **Delete** by note number

DATE FORMAT
`YYYY-MM-DD`

NOTE NUMBERS
Start from 1

SEARCH
Ignores **CAPITAL** / small letters

2 Tools and concepts used

ENVIRONMENT

Python 3 + VS Code

I wrote the program in Python 3 using Visual Studio Code.

STRINGS

Text handling

`strip()` removes extra spaces, `lower()` and `find()` are used for searching, and `split()` separates the date from the text.

FILES

Saving and reading

`open()` saves notes and reads them back, with the modes `"a"` (add), `"r"` (read) and `"w"` (write).

FUNCTIONS

One function per job

I made a separate function for each job, so the code is easier to read.

LISTS

Notes in memory

I keep the notes in a list while the program is working on them.

TRY AND EXCEPT

No crashes

I used these so the program shows a message instead of crashing when the input is wrong.

3 How my program works

`load_notes()`

Reads **diary.txt** and gives back all the notes as a list. It skips empty lines. If the file does not exist yet, it gives back an empty list, so the program does not crash. I made this function because view, search and delete all needed the same code, and I did not want to write it three times.

`add_note()`

Asks for the date and the text. First it checks the date with `datetime.strptime()`. Then it checks that the text is not empty. If everything is fine, the note is added at the end of diary.txt and the program prints **“Note saved.”**

`view_notes()`

Shows all the notes with numbers starting from 1. If there are no notes, it prints **“No notes found. Your diary is empty.”**

`search_notes()`

Asks for a keyword and turns both the keyword and the note text into small letters, so “Python” and “python” match. It uses `find()` to check whether the keyword is inside the note. The matching notes are shown with their original numbers.

`delete_note()`

Asks for a note number. If the number is wrong, it shows a message. If it is right, the note is removed from the list and all the remaining notes are written back to the file.

`Main menu`

A **while** loop keeps showing the menu until the user chooses **5** to exit. Each choice runs the matching function.

4 How the notes are stored

Each note is **one line** in `diary.txt`, with the date and text separated by a "|" sign, like this:

2026-10-10	|	Started Python project
DATE	SEPARATOR	TEXT

When I read a note back, I use `split("|", 1)`. The **1** means it cuts only at the **first** "|". I did this because a user might type a "|" inside the note itself, for example **"Buy milk | bread"**, and I did not want the text to get cut.

5 Problems I faced

`UnboundLocalError`

WHAT HAPPENED

In `view_notes` I forgot to write `number = 1` before using it.

HOW I FIXED IT

I set the number before the loop starts.

`IndexError`

WHAT HAPPENED

A blank line was left in diary.txt after I cleared it, and the program tried to split it.

HOW I FIXED IT

I made the program skip empty lines.

`Wrong note deleted`

WHAT HAPPENED

I once deleted the wrong note by typing 2 instead of 3.

WHAT I LEARNED

It made me understand how delete works.

These mistakes helped me understand error messages and testing better.

6 Test cases

4 / 4

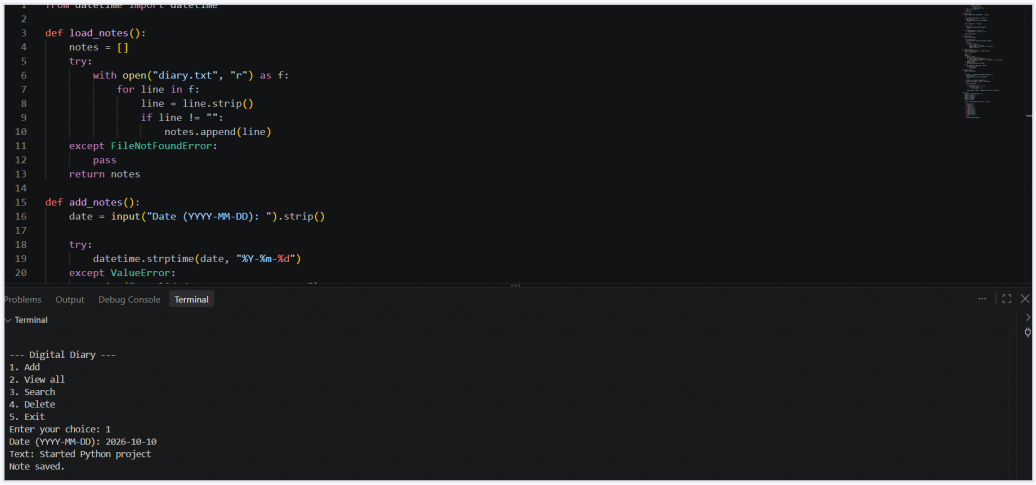
I ran all four test cases from the project sheet. In every case, the output of my program matched the expected output.

TEST CASE 1

Normal input

MATCHED

I chose option 1 and added the date 2026-10-10 with the text “Started Python project”. The program printed **“Note saved.”** Then option 2 showed `1 | 2026-10-10 | Started Python project`.



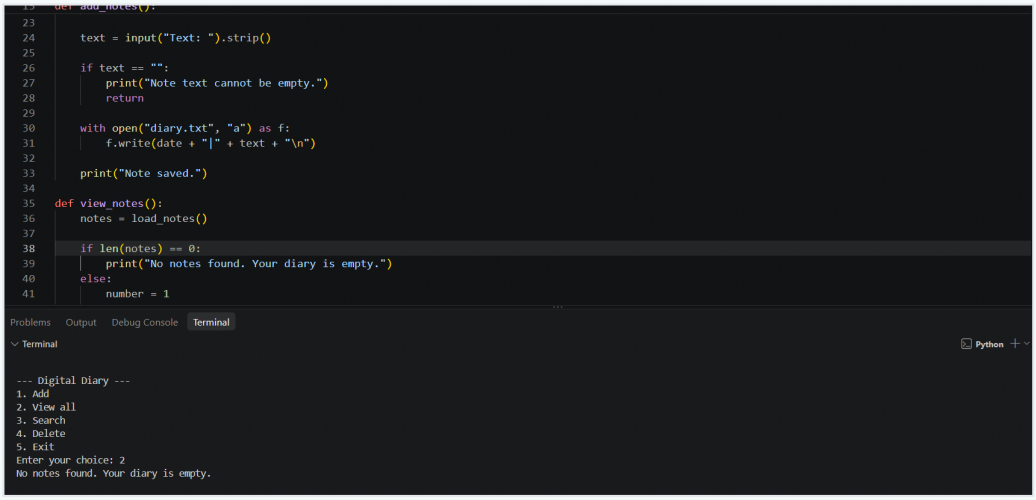
Screenshot: adding a note

TEST CASE 2

Empty diary

MATCHED

With an empty diary, I chose option 2. The program printed **“No notes found. Your diary is empty.”**



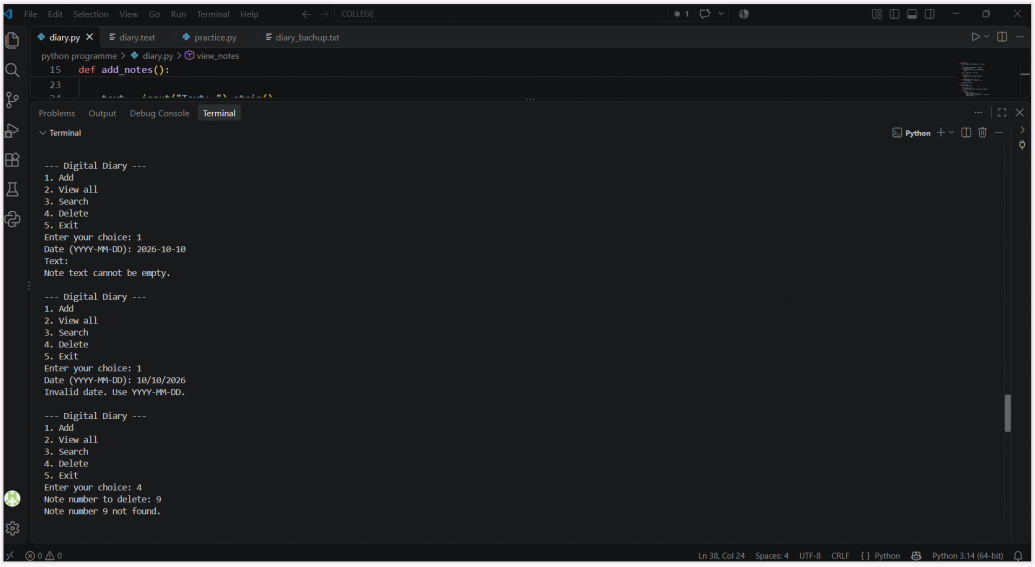
Screenshot: viewing an empty diary

TEST CASE 3

Invalid input

MATCHED

I tested three wrong inputs: **(a)** adding a note with empty text, which gave “Note text cannot be empty.”; **(b)** adding a note with the date 10/10/2026, which gave “Invalid date. Use YYYY-MM-DD.”; and **(c)** deleting note number 9, which gave “Note number 9 not found.”



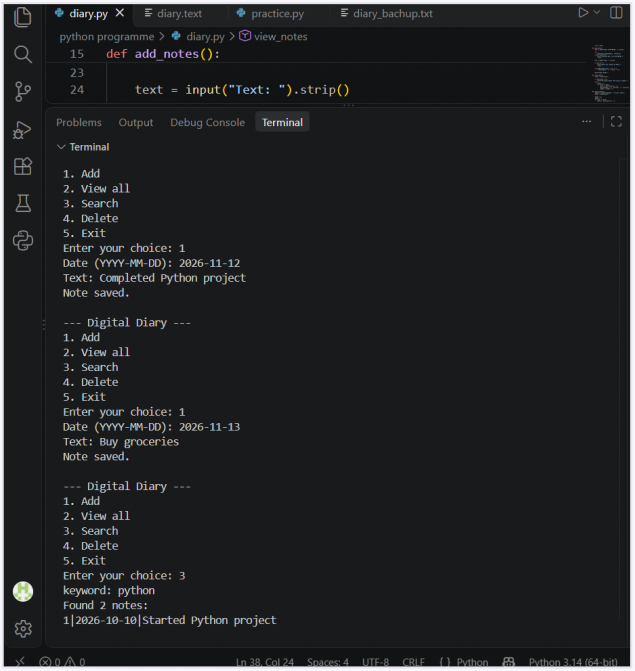
Screenshot: the three wrong inputs

TEST CASE 4

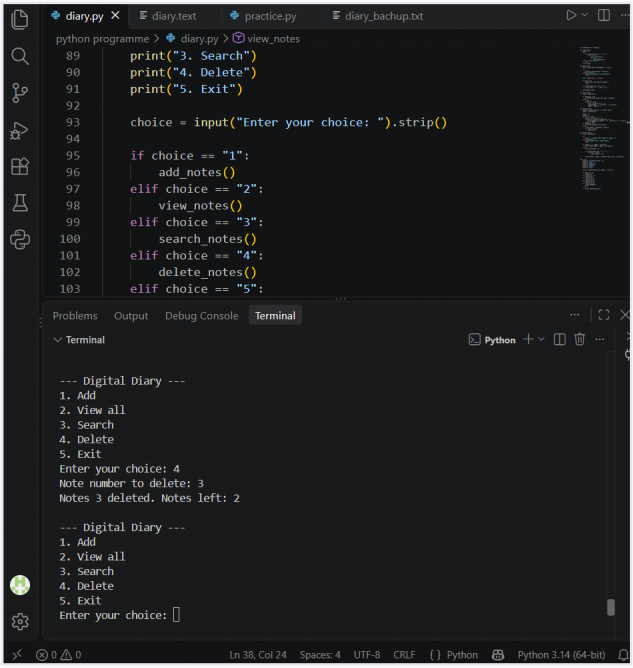
Different valid scenario

MATCHED

The diary already had the note 2026-10-10 “Started Python project” from Test Case 1. I added two more notes: 2026-11-12 “Completed Python project” and 2026-11-13 “Buy groceries”, so there were three notes in total. **(a)** Searching for “python” found 2 notes, notes 1 and 2. **(b)** Deleting note 3 printed “Note 3 deleted. Notes left: 2”.



(a) Adding notes and searching for “python”



(b) Deleting note 3

7 Input checks

My program checks these wrong inputs and shows a message instead of crashing:

WRONG INPUT
Empty note text
WHAT THE PROGRAM DOES
`"Note text cannot be empty."`

WRONG INPUT
Wrong date format (10/10/2026) or a date that does not exist (2026-13-45)
WHAT THE PROGRAM DOES
`"Invalid date. Use YYYY-MM-DD."`

WRONG INPUT
A note number that does not exist
WHAT THE PROGRAM DOES
`"Note number 9 not found."`

WRONG INPUT
Letters instead of a number when deleting
WHAT THE PROGRAM DOES
`"Please enter a valid note number."`

WRONG INPUT
A missing diary.txt file
WHAT THE PROGRAM DOES
The program treats it as an empty diary.

WRONG INPUT
A "|" sign inside the note text
WHAT THE PROGRAM DOES
The note is still shown completely.

8 Conclusion

This project helped me understand how a program can save data to a file and read it back. I learned how to use functions, lists, string methods and try/except. Testing the program with different inputs showed me that I must think about wrong inputs too, not only correct ones.

WHAT I LEARNED
- Saving data to a file and reading it back
- Functions and lists
- String methods
- try / except
- Testing with wrong inputs

IN THE FUTURE I COULD ADD
- A way to **edit** a note
- A way to **sort** the notes by date

9 Code

diary.py

```python
from datetime import datetime


def load_notes():
    notes = []
    try:
        with open("diary.txt", "r") as f:
            for line in f:
                line = line.strip()
                if line != "":
                    notes.append(line)
    except FileNotFoundError:
        pass
    return notes


def add_note():
    date = input("Date (YYYY-MM-DD): ").strip()

    try:
        datetime.strptime(date, "%Y-%m-%d")
    except ValueError:
        print("Invalid date. Use YYYY-MM-DD.")
        return

    text = input("Text: ").strip()

    if text == "":
        print("Note text cannot be empty.")
        return

    with open("diary.txt", "a") as f:
        f.write(date + "|" + text + "\n")

    print("Note saved.")


def view_notes():
    notes = load_notes()

    if len(notes) == 0:
        print("No notes found. Your diary is empty.")
    else:
        number = 1
        for note in notes:
            parts = note.split("|", 1)
            print(number, "|", parts[0], "|", parts[1])
            number = number + 1


def search_notes():
    keyword = input("Keyword: ").strip().lower()
    notes = load_notes()
    found = []
    number = 1

    for note in notes:
        parts = note.split("|", 1)
        if parts[1].lower().find(keyword) != -1:
            found.append(str(number) + " | " + parts[0] + " | " + parts[1])
        number = number + 1
```

```
    if len(found) == 0:
        print("No matching notes found.")
    else:
        print("Found", len(found), "notes:")
        for item in found:
            print(item)


def delete_note():
    notes = load_notes()

    try:
        number = int(input("Note number to delete: "))
    except ValueError:
        print("Please enter a valid note number.")
        return

    if number < 1 or number > len(notes):
        print("Note number", number, "not found.")
    else:
        notes.pop(number - 1)
        with open("diary.txt", "w") as f:
            for note in notes:
                f.write(note + "\n")
        print("Note", number, "deleted. Notes left:", len(notes))


# ----- Main menu -----
while True:
    print("\n--- Digital Diary ---")
    print("1. Add")
    print("2. View all")
    print("3. Search")
    print("4. Delete")
    print("5. Exit")

    choice = input("Enter your choice: ").strip()

    if choice == "1":
        add_note()
    elif choice == "2":
        view_notes()
    elif choice == "3":
        search_notes()
    elif choice == "4":
        delete_note()
    elif choice == "5":
        print("Goodbye!")
        break
    else:
        print("Invalid choice")

```